

Solved selected problems of Information Theory, Inference, and Learning Algorithms by David J.C. MacKay.

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1 - Introduction to Information Theory

Solution. **Exercise 1.2**

When using R_3 we get an error if two or three bits are flipped in a set of 3 bits. Then the probability of two bits being flipped is

$$P(r = 2 \mid f, N = 3) = \binom{3}{2} f^2 (1 - f)^{3-2} = 3f^2(1 - f)$$

And the probability of three bits being flipped is

$$P(r = 3 \mid f, N = 3) = \binom{3}{3} f^3 (1 - f)^{3-3} = f^3$$

Then the total probability of getting an error when using R_3 is

$$P(2 \mid f, 3) + P(3 \mid f, 3) = f^3 + 3f^2(1 - f) = 3f^2 - 2f^3$$

We observe that $3f^2 - 2f^3$ is less than f if $f < 0.5$, hence, the error probability is reduced by the use of R_3 . $\square$

Solution. **Exercise 1.3**

- (a) Let us consider the repetition code R_N where N is odd.

We observe that we get an error when recovering the bits if at least $(N + 1)/2$ bits are flipped, the probability of this happening is

$$P((N + 1)/2 \mid f, N) = \binom{N}{(N + 1)/2} f^{(N+1)/2} (1 - f)^{N-(N+1)/2}$$

If $(N + 1)/2 + 1$ or $(N + 1)/2 + 2$ (and so on) bits are flipped then we also get errors, so we need to add up all the probabilities of each of these cases occurring, therefore we get that the error probability is

$$p_b = \sum_{n=(N+1)/2}^N \binom{N}{n} f^n (1 - f)^{N-n}$$

- (b) Let $f = 0.1$ and let us consider the ratio of two subsequent terms

$$\frac{f^n (1 - f)^{N-n}}{f^{n+1} (1 - f)^{N-n-1}} = \frac{(1 - f)}{f} = 9$$

This implies that $f^n (1 - f)^{N-n} > f^{n+1} (1 - f)^{N-n-1}$ for any n , so the biggest term is the one with the lowest n i.e. $n = (N + 1)/2$.

The second-biggest term is then $(N + 1)/2 + 1$ and we saw that the relation between subsequent terms for $f = 0.1$ is

$$f^n (1 - f)^{N-n} = 9 \cdot f^{n+1} (1 - f)^{N-n-1}$$

So the biggest term is 9 times bigger than the second-biggest term.

- (c) The largest term happens when $n = (N + 1)/2$ so by Stirling's approximation the largest term is given by

$$\begin{aligned} \binom{N}{(N + 1)/2} f^{(N+1)/2} (1 - f)^{(N-1)/2} &= \\ &= 2^{NH_2((N+1)/2N)} f^{(N+1)/2} (1 - f)^{(N-1)/2} \end{aligned}$$

Since this term is 9 times the next one then we may assume that the probability of error is

$$p_B \approx 2^{NH_2((N+1)/2N)} f^{(N+1)/2} (1 - f)^{(N-1)/2}$$

(d) If N is reasonably big, we may approximate $H_2((N+1)/2N)$ as

$$H_2((N+1)/2N) = H_2(1/2 + 1/2N) \approx H_2(1/2) = 1$$

And hence the probability of error becomes

$$p_B \approx 2^N f^{N/2} (1-f)^{N-N/2} = 2^N (f(1-f))^{N/2}$$

We want that $p_B \leq 10^{-15}$ so taking logarithms on both sides and solving for N we get that

$$\begin{aligned} N + N \log(\sqrt{f(1-f)}) &= \log(10^{-15}) \\ N &= \frac{\log(10^{-15})}{1 + \log(\sqrt{f(1-f)})} \end{aligned}$$

If we plug in $f = 0.1$ then we get that $N \approx 67$.

□

Solution. **Exercise 1.4**

We note that

$$\mathbf{H}\mathbf{G}^T = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Since all the codewords are determined by $\mathbf{t} = \mathbf{G}^T \mathbf{s}$ then

$$\mathbf{H}\mathbf{t} = \mathbf{H}\mathbf{G}^T \mathbf{s} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

□

Solution. **Exercise 1.5**

- (a) The received message is $\mathbf{r} = 1101011$ we note that

$$r_1 + r_2 + r_3 + r_5 = 1 + 1 + 0 + 0 = 0 \Rightarrow \text{"happy circle"}$$

$$r_2 + r_3 + r_4 + r_6 = 1 + 0 + 1 + 1 = 1 \Rightarrow \text{"unhappy circle"}$$

$$r_1 + r_3 + r_4 + r_7 = 1 + 0 + 1 + 1 = 1 \Rightarrow \text{"unhappy circle"}$$

The only received bit inside the two "unhappy" circles is r_4 so it must be flipped, and hence the decoded message must be 1100011. Using Algorithm 1.16 since the syndrome is $\mathbf{z} = 011$ tells us also that r_4 must be flipped.

- (b) The received message is $\mathbf{r} = 0110110$ we note that

$$r_1 + r_2 + r_3 + r_5 = 0 + 1 + 1 + 1 = 1 \Rightarrow \text{"unhappy circle"}$$

$$r_2 + r_3 + r_4 + r_6 = 1 + 1 + 0 + 1 = 1 \Rightarrow \text{"unhappy circle"}$$

$$r_1 + r_3 + r_4 + r_7 = 0 + 1 + 0 + 0 = 0 \Rightarrow \text{"happy circle"}$$

The syndrome is $\mathbf{z} = 110$ so using Algorithm 1.16, r_2 must be flipped and hence the decoded message is 0010110.

- (c) The received message is $\mathbf{r} = 0100111$ we note that

$$r_1 + r_2 + r_3 + r_5 = 0 + 1 + 0 + 1 = 0 \Rightarrow \text{"happy circle"}$$

$$r_2 + r_3 + r_4 + r_6 = 1 + 0 + 0 + 1 = 0 \Rightarrow \text{"happy circle"}$$

$$r_1 + r_3 + r_4 + r_7 = 0 + 0 + 0 + 1 = 1 \Rightarrow \text{"unhappy circle"}$$

The syndrome is $\mathbf{z} = 001$ so using Algorithm 1.16, r_7 must be flipped and hence the decoded message is 0100110.

- (d) The received message is $\mathbf{r} = 1111111$ we note that

$$r_1 + r_2 + r_3 + r_5 = 1 + 1 + 1 + 1 = 0 \Rightarrow \text{"happy circle"}$$

$$r_2 + r_3 + r_4 + r_6 = 1 + 1 + 1 + 1 = 0 \Rightarrow \text{"happy circle"}$$

$$r_1 + r_3 + r_4 + r_7 = 1 + 1 + 1 + 1 = 0 \Rightarrow \text{"happy circle"}$$

The syndrome is $\mathbf{z} = 000$ so there is no flipped bit and the decoded message is the same 1111111.

□

Solution. **Exercise 1.6**

- (a) To compute the probability of block error p_B we compute the probability of a message of 7 bits to have 2 errors, where the Hamming decoding algorithm would not be able to decode it correctly.

$$\begin{aligned} P(2 \mid f, 7) &= \binom{7}{2} f^2 (1-f)^{7-2} \\ &= 21 f^2 (1-f)^5 \\ &= 21 f^2 - 105 f^3 + 210 f^4 - 210 f^5 + 105 f^6 - 21 f^7 \end{aligned}$$

If we compute the probability of a message to have 3 bits flipped we get a leading term of the order f^3 , which is going to be smaller to the leading term of f^2 in the case of 2 bits flipped, as we saw. The same thing happens for the probabilities of having 4, 5, 6, and 7 bits flipped, where the leading term is smaller than the f^2 leading term we got. Therefore, the leading order is $21 f^2$.

- (b) When we have 2 errors in a message then the decoder will try to fix the message by flipping another bit so we end up with 3 errors after decoding.

If we assume that the probability of block error is the probability of the whole block having an error then $3p_B/7$ is approximately the probability of having 3 errors out of 7 bits, then taking only the leading term we see that the probability of bit error is

$$p_b \approx \frac{3}{7} \cdot 21 f^2 = 9 f^2$$

□

Solution. **Exercise 1.7**

We know the noise vector $\mathbf{n}$ satisfies $\mathbf{H}\mathbf{n} = \mathbf{z}$ where $\mathbf{H}$ is the parity check matrix and $\mathbf{z}$ is the syndrome vector. We are interested in some $\mathbf{n}$ such that

$$\mathbf{H}\mathbf{n} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

But also we proved that

$$\mathbf{H}\mathbf{t} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

where $\mathbf{t}$ is a codeword vector.

Therefore, setting $\mathbf{n} = \mathbf{t}$ we get a noise vector that give all-zero syndrome. Since there are 16 codewords in a (7,4) Hamming code then there are 16 noise vectors that satisfy this. $\square$

Solution. **Exercise 1.8**

If two bit flips happened then the decoder will try to fix it assuming there is only one bit flipped and hence will try to flip some bit.

Two things may happen, if the decoder decides to flip one of the wrong bits then we still have one error after decoding.

On the other hand, if the decoder decides to flip a different bit then after decoding we have three errors.

So, either way we have a block decoding error.

If more than two bits flipped it's even worst because the Hamming decoder can only flip 1 bit so after the decoding process we still have an error. $\square$

Solution. **Exercise 1.12**

The probability of getting an error when using R_3 is $3f^2 - 2f^3$ so if we use it twice to decode a message again we get a probability of error of

$$p_b(R_3^2) = (3f^2 - 2f^3)(3f^2 - 2f^3) = 9f^4 - 12f^5 + 4f^6$$

On the other hand, the probability of error of R_9 is

$$p_b(R_9) = \sum_{n=5}^9 \binom{9}{n} f^n (1-f)^{9-n}$$

We see that the leading term is of the order f^5 so the probability of error of R_9 is smaller than the probability of error of R_3^2 which leading term is of the order f^4 . $\square$